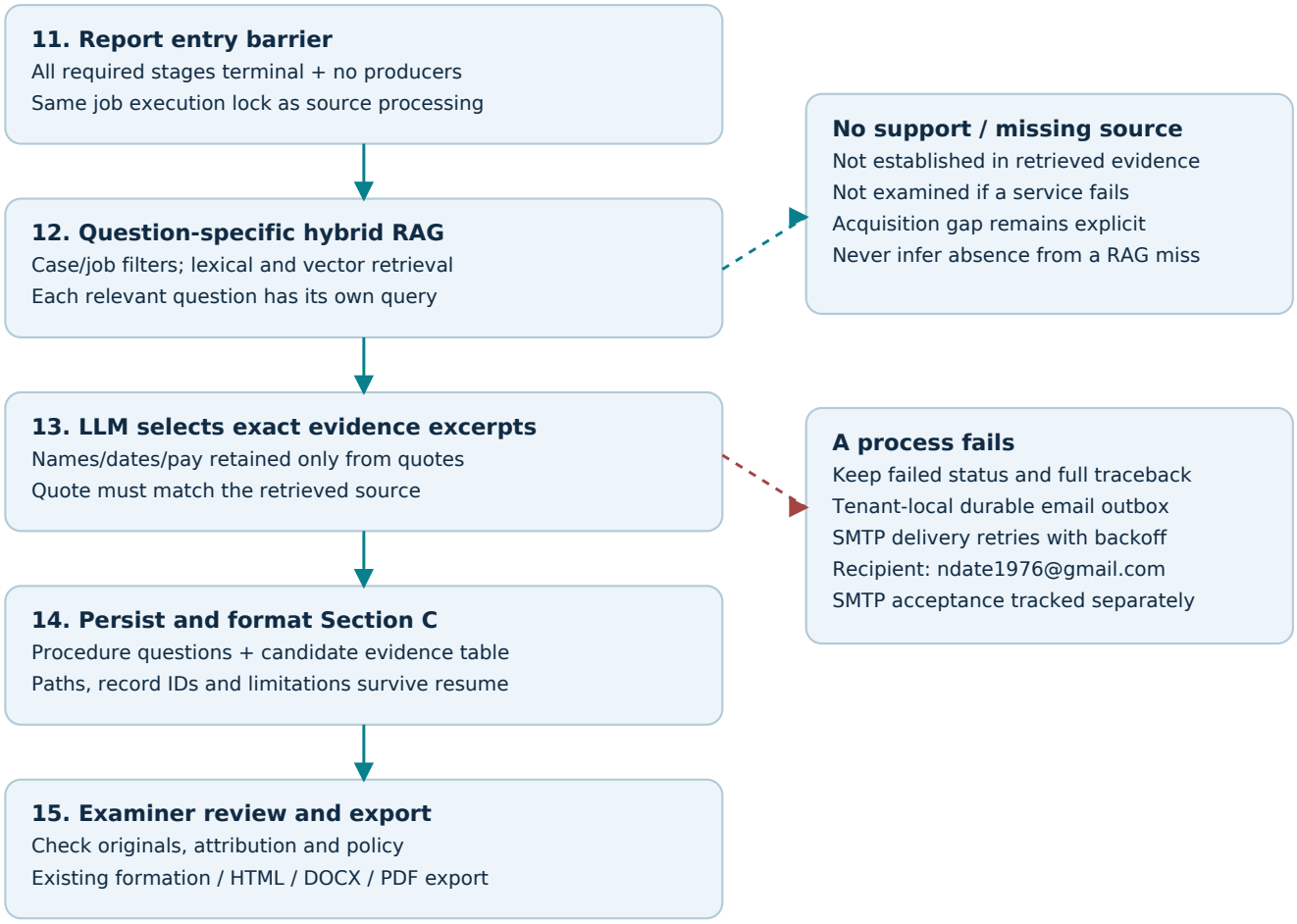
Android image to sealed evidence

The diagram maps the existing engines and the V78 completion boundaries. The stage controller advances one dependent stage at a time; case-scoped CPU and GPU helpers may overlap only inside the marked windows.



All stages retain job and tenant scope. Allocated, deleted, carved and partial records remain distinguishable. No key is derived from a CRYPT backup, and deleted data is recovered only where evidence permits.

Sealed evidence to reviewed report



Output contract

The deterministic observation remains authoritative for catalog counts. Supplemental findings are candidate excerpts requiring examiner review. A document on a device is not proof of sending it; an offer is not proof of acceptance; an app installation is not proof of use; suspicion is not a concluded policy violation.

State in UI	Meaning
wait	Dependency or resource is not ready.
next	First eligible stage queued or ready for delivery.
in process	Worker is processing evidence below its denominator.
writing in	Database-writing stage, finalization, or a producer barrier remains open.
completed	Controller received the worker result and passed committed database barriers.
failed	A process failed; traceback is retained and delivery is queued.

Internal database statuses remain compatible with the dispatcher. The six requested labels are exposed as process_status and displayed by the forensic pipeline UI. Paused controls and documented skipped/failed evidence exceptions remain visible.

Performance and concurrency review

Finding in the upload	Change / retained design
Read-merge-write progress could lose concurrent fields.	Lock the job row before merging. Unrelated jobs still proceed independently.
99% coordinator could seal while the producer was alive.	Remove count-only closure; require returned worker result plus barriers. OCR sidecar 100% cannot seal the authoritative OCR stage.
Sidecar claims were read then overwritten.	Use atomic conditional claim and per-sidecar execution lock; obsolete deliveries do no work.
Sidecar final writes could overlap authoritative output.	Drain producers at inventory, chunk and report boundaries. Report holds the same case execution lock as source stages.
Quality triage could refill an OCR bucket late.	Serialize quality producers; late tasks stop once OCR is sealed. Triage errors hold the OCR stage.
Graph expansion could traverse shared catalog nodes.	Anchor traversal to case-owned evidence and constrain every traversed node to the case.
One broad query could starve additional questions.	Independent query for each question; report-local query cache scoped by tenant/job/question. At most two short candidates per question.
Completeness anti-joins can be costly over many chunks.	Optional online indexes on job_id, mobile/priority artifact identifier and chunk policy. Use the included installer before measuring plans.
Unbounded resource admission risks contention.	Existing bounded file workers retained; independent helper admission now fails closed when capacity checks are unavailable.

Retain / remove / add

Retain immutable source evidence, finalized manifests, SQLite/WAL grouping, bounded buffers, idempotent record identities, OCR quality gating, hybrid retrieval and deterministic counts. Remove count-only auto-completion and unscoped graph walks. Add producer joins, atomic claims, full-traceback outbox and quote-validated investigative findings.

What still needs measurement on your system

No production image, PostgreSQL server, CUDA model service or SMTP server was executed here. Do not treat code review as a files/minute or accuracy benchmark. Measure wall time per stage, rows written per second, peak RAM, CPU/GPU use, database wait time and buffer reads. Compare same evidence hashes and row identities before and after the release. Do not shorten parser coverage or OCR scope to improve throughput.

Use EXPLAIN (ANALYZE, BUFFERS) on a copy of the case for the mobile/priority chunk anti-joins and pending-work queries. Tune workers against available PostgreSQL connections and storage throughput; several helper slots do not create extra database capacity.

Investigation questions 1-8

Questions are added to matching Section C procedures and assessed through case-scoped retrieval. Apply the consent, incident period, organization policy and custodian scope before concluding a finding.

01. Offer

If the custodian is an employee, is there an offer letter or recruitment communication? Identify employer, role, issue/joining dates, remuneration and channel where evidenced.

02. Outside Work

Is there evidence of outside employment, consulting, competing business, or recruitment discussions? Separate proposals from accepted employment.

03. Chat Conduct

Do conversations contain case-relevant requests to conceal activity, share company material, bypass controls, or coordinate an unauthorized action? Examine full conversational context.

04. Document Transfer

Were confidential company documents sent through chat, email or another channel? Identify sender, recipient/group, timestamp, attachment, direction and delivery evidence; presence alone is not transfer.

05. Company Images

Do screenshots, photographs or videos show company documents, source code, customer data, credentials, restricted premises or screen recordings? Require examiner review of the original media.

06. Personal Destination

Was company information directed to personal accounts, external recipients, public groups, shared links or cloud storage? Establish destination and account attribution.

07. Classification

What evidence or supplied company policy establishes that the material is confidential and that the custodian lacked authorization?

08. Bulk Copy

Are there bulk copy, archive, export, USB/OTG or upload events? Correlate filenames, hashes, times and destination records.

Investigation questions 9-16

Questions are added to matching Section C procedures and assessed through case-scoped retrieval. Apply the consent, incident period, organization policy and custodian scope before concluding a finding.

09. Deletion

Are there relevant deletion, wiping, disappearing-message, reset or cleanup traces? Distinguish recovered remnants from intact records and do not infer intent from deletion alone.

10. Accounts

Which accounts, SIMs, profiles and identifiers can be attributed to this device or custodian? Record attribution uncertainty.

11. Timeline

What is the supported event sequence around the incident, recruitment, transfer or departure date? Preserve original timestamps and timezone uncertainty.

12. Contacts

Are communications with named competitors, recruiters, vendors or unusual counterparties supported? Contact presence alone does not establish a relationship.

13. Browser

Are relevant job portals, webmail, file-sharing, phishing or data-upload pages evidenced? Separate visits, downloads and uploads.

14. Apps

Were remote-access, alternate messaging, vault, cloning, rooting or sideloaded apps used in a case-relevant way? Installation alone does not establish use or wrongdoing.

15. Network

Are relevant Wi-Fi, VPN, Bluetooth or hotspot connections supported and correlated with the event timeline?

16. Location

Do GPS, EXIF or location records support a case-relevant place/time? Do not treat isolated coordinates as proof of a person’s presence.

Investigation questions 17-22

Questions are added to matching Section C procedures and assessed through case-scoped retrieval. Apply the consent, incident period, organization policy and custodian scope before concluding a finding.

17. Financial

Are case-relevant payment requests, benefits or financial inducements linked to company information? Separate discussion from an evidenced transaction.

18. Malware

Are phishing, credential theft, malicious applications or suspicious permission changes directly supported? Distinguish a suspicious indicator from confirmed compromise.

19. Auth

Are unusual logins, account changes, MFA changes or sharing of credentials directly supported?

20. Recovery

Which deleted records are recoverable, and which sources are encrypted, overwritten, inaccessible or absent? Preserve recovery provenance and acquisition limitations.

21. Corroboration

Can the candidate event be corroborated across independent sources or matching hashes? Identify contradictions and benign explanations.

22. Scope

Which acquisition scope, integrity hashes, tool versions, source limitations and time settings constrain the examination?

Evidence fields and review

For offer letters: employer, role, issue date, joining date, remuneration and channel, when literally evidenced. For transfers: sender, recipient/group, timestamp, direction, attachment and channel. Each candidate retains its exact quote, source path and record/chunk identifier. Unknown fields are omitted, never completed by inference.

A review table records the question, result and supporting source. Original context, competing explanations, account attribution, recovery state, timezone and the actual confidentiality/authorization policy require examiner verification. The 22 questions are an extensible case playbook, not a claim to cover every forensic investigation.

Deployment and acceptance checks

Use the compose file for the stack already hosting your case. Preserve the existing environment, certificates, evidence and database volumes. This release adds tenant-local control tables automatically; the online evidence indexes are installed separately.

Root forensic stack - PowerShell, from project root

```
docker compose build --no-cache api frontend

docker compose up -d --no-build

docker compose exec -T api python scripts/install_process_indexes_v78.py
```

Separate Android stack - PowerShell, from project root

Set a command prefix variable, then rebuild only the shared API image and frontend. All Android workers reuse the backend image.

```
$dcArgs = @('-f','services/mobile-android/docker-compose.yml',

            '--project-directory','.')

docker compose @dcArgs build --no-cache api frontend

docker compose @dcArgs up -d --no-build

docker compose @dcArgs exec -T api python scripts/install_process_indexes_v78.py
```

Failure email configuration

Use the existing MAIL_HOST, MAIL_PORT, MAIL_USERNAME, MAIL_PASSWORD, MAIL_FROM, MAIL_SMTP_AUTH, MAIL_SMTP_STARTTLS and MAIL_SMTP_SSL settings. The default recipient is ndate1976@gmail.com. FORENSIC_FAILURE_EMAIL can change it; FORENSIC_FAILURE_MAIL_ENABLED=false pauses alert delivery. Alert mail has its own switch so mobile workers may still disable identity mail. These variables must reach the worker environment.

Keep worker-progress and worker-beat running with PIPELINE_SUPERVISOR_ENABLED enabled. Full child-process tracebacks are retained in the outbox and then emailed. If SMTP is unavailable, the failed processing state remains correct and the alert retries. Delivery is at least once: an ambiguous SMTP response or a crash after acceptance can duplicate an alert. A hard-killed process may leave no Python traceback; the alert reports the available failure/retry context.

On your deployment, verify before processing a real case

Run two synthetic cases concurrently and redeliver a task. Confirm tenant/job separation, identical semantic record identities, no duplicate sidecar execution, and that reports stay held while any required writer is active. Inject an exception with a nested traceback and verify the outbox, then SMTP acceptance at the configured recipient. Check missing-key, deleted-record and empty-source limitations in the report. Regenerate reports to include the new questions; rerun affected processing when earlier completion was premature.

Validation record and practical limits

Verified here

The expanded focused regression set passed 99 tests. The final release verification is recorded in V78_TEST_RESULTS.txt inside the project. Checks cover status mapping, producer barriers, quote validation, artifact identity isolation, full remote trace preservation, SMTP retry state, retrieval scope, graph scope, OCR overlap, mobile inventory and shared-image build reuse. The Android React/TypeScript production build passed. Python compilation passed.

Original-upload test debt

A separate baseline run reproduced older failures: fixed four-worker expectations against the current configurable cap, materialization mocks that omit the WhatsApp gate, and old procedure wording expectations. One four-thread barrier test was also nondeterministic under the wider worker cap. These are recorded separately from the focused release checks; they do not establish that every existing test passes.

Limits of this release

No real evidence image was processed here. PostgreSQL concurrency and query plans, live SMTP delivery, GPU/OCR throughput, report export pagination with an actual case and end-to-end forensic accuracy remain deployment acceptance checks. Citation/exact-quote checks prevent invented excerpts; they do not prove that a selected excerpt is authentic, complete or relevant without examiner review.

The requested changes target Android mobile and shared Disk/Mobile control/report paths. Existing vulnerability scanner engines and scoring semantics are retained. This release does not claim AXIOM or Nessus feature or detection parity.

Primary technical references

PostgreSQL 18: explicit locking - https://www.postgresql.org/docs/18/explicit-locking.html

Python: traceback formatting - https://docs.python.org/3/library/traceback.html

Python: SMTP client - https://docs.python.org/3/library/smtplib.html

Prepared from the supplied polaron(10).zip on 9 October 2026. Source originals and existing project configuration are retained; added controls operate on derived records and process state.